

CHAPTER

5.7

THE CONTINUOUS-TIME FOURIER SERIES

Statement for Q.1-5:

Determine the Fourier series coefficient for given periodic signal $x(t)$.

1. $x(t)$ as shown in fig. P5.7.1

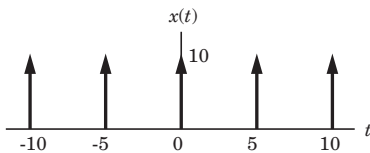


Fig. P5.7.1

- (A) 1
(B) $\cos\left(\frac{\pi}{2}k\right)$
(C) $\sin\left(\frac{\pi}{2}k\right)$
(D) 2

2. $x(t)$ as shown in fig. P5.7.2

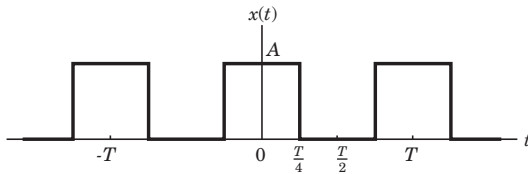


Fig. P5.7.2

- (A) $\frac{A}{j\pi k} \sin\left(\frac{\pi}{2}k\right)$
(B) $\frac{A}{j\pi k} \cos\left(\frac{\pi}{2}k\right)$
(C) $\frac{A}{\pi k} \sin\left(\frac{\pi}{2}k\right)$
(D) $\frac{A}{\pi k} \cos\left(\frac{\pi}{2}k\right)$

3. $x(t)$ as shown in fig. P5.7.3

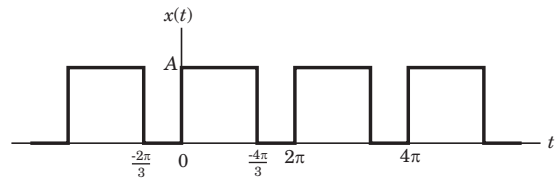


Fig. P5.7.3

- (A) $\frac{A}{2\pi k} \left(e^{-j\left(\frac{4\pi k}{3}\right)} - 1 \right)$
(B) $j \frac{A}{2\pi k} \left(e^{-j\left(\frac{4\pi k}{3}\right)} - 1 \right)$
(C) $-j \frac{A}{2\pi k} \left(e^{-j\left(\frac{4\pi k}{3}\right)} - 1 \right)$
(D) $\frac{-A}{2\pi k} \left(e^{-j\left(\frac{4\pi k}{3}\right)} - 1 \right)$

4. $x(t)$ as shown in fig. P5.7.4

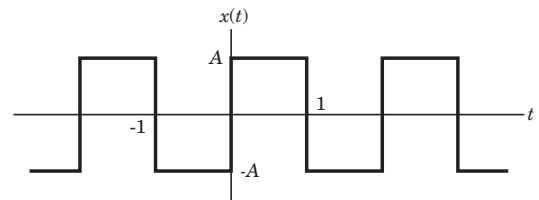


Fig. P5.7.4

- (A) $\frac{A}{k\pi} (1 - (-1)^k)$
(B) $\frac{A}{k\pi} (1 + (-1)^k)$
(C) $\frac{A}{jk\pi} (1 - (-1)^k)$
(D) $\frac{A}{jk\pi} (1 + (-1)^k)$

5. $x(t) = \sin^2 t$

- (A) $-\frac{1}{4} \delta[k-1] + \frac{1}{2} \delta[k] - \frac{1}{4} \delta[k+1]$
(B) $-\frac{1}{4} \delta[k-2] + \frac{1}{2} \delta[k] - \frac{1}{4} \delta[k+2]$
(C) $-\frac{1}{2} \delta[k-1] + \delta[k] - \frac{1}{2} \delta[k+1]$
(D) $-\frac{1}{2} \delta[k-2] + \delta[k] - \frac{1}{2} \delta[k+2]$

Statement for Q.6-11:

In the question, the FS coefficient of time-domain signal have been given. Determine the corresponding time domain signal and choose correct option.

7. $X[k] = j\delta[k-1] - j\delta[k+1] + \delta[k+3] + \delta[k-3]$, $\omega_0 = 2\pi$

- (A) $2(\cos 3\pi t - \sin \pi t)$ (B) $-2(\cos 3\pi t - \sin \pi t)$
(C) $2(\cos 6\pi t - \sin 2\pi t)$ (D) $-2(\cos 6\pi t - \sin 2\pi t)$

8. $X[k] = \left(\frac{-1}{3}\right)^{|k|}$, $\omega_0 = 1$

- (A) $\frac{4}{5 + 3\sin t}$ (B) $\frac{5}{4 + 3\sin t}$
(C) $\frac{5}{4 + 3\cos t}$ (D) $\frac{4}{5 + 3\cos t}$

9. $X[k]$ as shown in fig. P5.7.9, $\omega_0 = \pi$

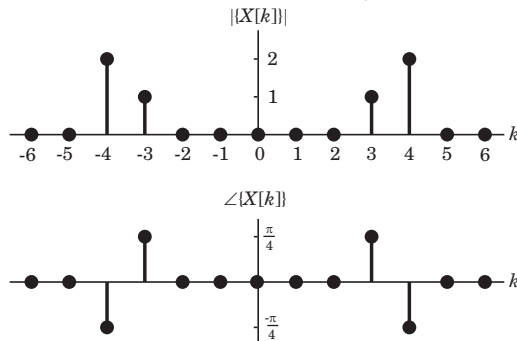


Fig. P5.7.9

- (A) $6 \cos \left(2\pi t + \frac{\pi}{4} \right) - 3 \cos \left(3\pi t - \frac{\pi}{4} \right)$
(B) $4 \cos \left(4\pi t - \frac{\pi}{4} \right) - 2 \cos \left(3\pi t + \frac{\pi}{4} \right)$
(C) $2 \cos \left(2\pi t + \frac{\pi}{4} \right) - 2 \cos \left(3\pi t - \frac{\pi}{4} \right)$
(D) $4 \cos \left(4\pi t + \frac{\pi}{4} \right) + 2 \cos \left(3\pi t - \frac{\pi}{4} \right)$

10. $X[k]$ As shown in fig. P5.7.10, $\omega_0 = 2\pi$

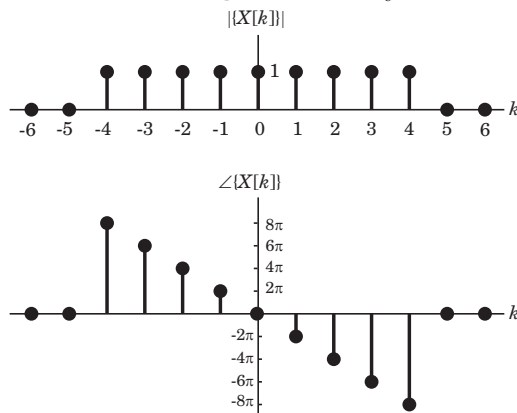


Fig. P5.7.10

(A) $\frac{\sin 9\pi t}{\sin \pi t}$

(B) $\frac{\sin 9\pi t}{\pi \sin \pi t}$

(C) $\frac{\sin 18\pi t}{2 \sin \pi t}$

(D) None of the above

11. $X[k]$ As depicted in fig. P5.7.11, $\omega_0 = \pi$

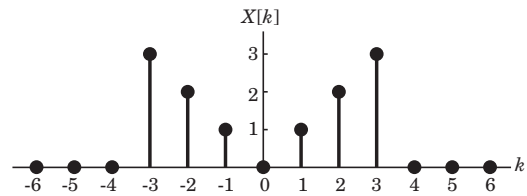


Fig. P5.7.11

- (A) $3 \cos 3\pi t + 2 \cos 2\pi t + \cos \pi t$
(B) $3 \sin 3\pi t + 2 \sin 2\pi t + \sin \pi t$
(C) $6 \sin 3\pi t + 4 \sin 2\pi t + 2 \sin \pi t$
(D) $6 \cos 3\pi t + 4 \cos 2\pi t + 2 \cos \pi t$

Statement for Q.12-16:

Consider a continuous time periodic signal $x(t)$ with fundamental period T and Fourier series coefficients $X[k]$. Determine the Fourier series coefficient of the signal $y(t)$ given in question and choose correct option.

12. $y(t) = x(t - t_0) + x(t + t_0)$

(A) $2 \cos \left(\frac{2\pi}{T} kt_0 \right) X[k]$

(B) $2 \sin \left(\frac{2\pi}{T} kt_0 \right) X[k]$

(C) $e^{-j\omega_0 t_0} X[k] + e^{j\omega_0 t_0} X[-k]$

(D) $e^{-j\omega_0 t_0} X[-k] + e^{j\omega_0 t_0} X[k]$

13. $y(t) = \text{Ev}\{x(t)\}$

(A) $\frac{X[k] + X[-k]}{2}$

(B) $\frac{X[k] - X[-k]}{2}$

(C) $\frac{X[k] + X^*[-k]}{2}$

(D) $\frac{X[k] + X^*[k]}{2}$

14. $y(t) = \text{Re}\{x(t)\}$

(A) $\frac{X[k] + X[-k]}{2}$

(B) $\frac{X[k] - X[-k]}{2}$

(C) $\frac{X[k] + X^*[-k]}{2}$

(D) $\frac{X[k] + X^*[k]}{2}$

15. $y(t) = \frac{d^2 x(t)}{dt^2}$

- (A) $\left(\frac{2\pi k}{T}\right)^2 X[k]$ (B) $-\left(\frac{2\pi k}{T}\right)^2 X[k]$
 (C) $j\left(\frac{2\pi k}{T}\right)^2 X[k]$ (D) $-j\left(\frac{2\pi k}{T}\right)^2 X[k]$

16. $y(t) = x(4t - 1)$

- (A) $\frac{8\pi}{T} X[k]$ (B) $\frac{4\pi}{T} X[k]$
 (C) $e^{-jk\left(\frac{8\pi}{T}\right)} X[k]$ (D) $e^{jk\left(\frac{8\pi}{T}\right)} X[k]$

17. Consider a continuous-time signal

$$x(t) = 4 \cos 100\pi t \sin 1000\pi t$$

with fundamental period $T = \frac{1}{50}$. The nonzero FS coefficient for this function are

- (A) $X[-4], X[4], X[-7], X[7]$
 (B) $X[-1], X[1], X[-10], X[10]$
 (C) $X[-3], X[3], X[-4], X[4]$
 (D) $X[-9], X[9], X[-11], X[11]$

18. A real valued continuous-time signal $x(t)$ has a fundamental period $T = 8$. The nonzero Fourier series coefficients for $x(t)$ are

$$X[1] = X[-1] = 4, X[3] = X^*[-3] = 4j$$

The signal $x(t)$ would be

- (A) $4 \cos\left(\frac{\pi}{4}t\right) + 4j \sin\left(\frac{3\pi}{4}t\right)$
 (B) $4 \cos\left(\frac{\pi}{4}t\right) - 4j \cos\left(\frac{3\pi}{4}t\right)$
 (C) $8 \cos\left(\frac{\pi}{4}t\right) + 8 \cos\left(\frac{3\pi}{4}t + \frac{\pi}{2}\right)$
 (D) None of the above

19. The continuous-time periodic signal is given as

$$x(t) = 4 + \cos\left(\frac{2\pi}{3}t\right) + 6 \sin\left(\frac{5\pi}{3}t\right)$$

The nonzero Fourier coefficients are

- (A) $X[0], X[-1], X[1], X[-5], X[5]$
 (B) $X[0], X[-2], X[2], X[-5], X[5]$
 (C) $X[0], X[-4], X[4], X[-10], X[10]$
 (D) None of the above

Statement for Q.20-21:

Let $x_1(t)$ and $x_2(t)$ be continuous time periodic signal with fundamental frequency ω_1 and ω_2 , Fourier series coefficients $X_1[k]$ and $X_2[k]$ respectively. Given that $x_2(t) = x_1(t - 1) + x_1(1 - t)$

20. The relation between ω_1 and ω_2 is

- (A) $\omega_2 = \frac{\omega_1}{2}$ (B) $\omega_2 = \omega_1^2$
 (C) $\omega_2 = \omega_1$ (D) $\omega_2 = \sqrt{\omega_1}$

21. The Fourier coefficient $X_2[k]$ will be

- (A) $(X_1[k] - jX_1[-k])e^{-j\omega_1 k}$
 (B) $(X_1[-k] - jX_1[k])e^{-j\omega_1 k}$
 (C) $(X_1[k] + jX_1[-k])e^{-j\omega_1 k}$
 (D) None of the above

Statement for Q.22-23:

Consider three continuous-time periodic signals whose Fourier series representation are as follows.

$$x_1(t) = \sum_{k=0}^{100} \left(\frac{1}{3}\right)^k e^{-jk\frac{2\pi}{50}t}$$

$$x_2(t) = \sum_{k=-100}^{100} \cos k\pi e^{-jk\frac{2\pi}{50}t}$$

$$x_3(t) = \sum_{k=-100}^{100} j \sin\left(\frac{k\pi}{2}\right) e^{-jk\frac{2\pi}{50}t}$$

22. The even signals are

- (A) $x_2(t)$ only (B) $x_2(t)$ and $x_3(t)$
 (C) $x_1(t)$ and $x_3(t)$ (D) $x_1(t)$ only

23. The real valued signals are

- (A) $x_1(t)$ and $x_2(t)$ (B) $x_2(t)$ and $x_3(t)$
 (C) $x_3(t)$ and $x_1(t)$ (D) $x_1(t)$ and $x_3(t)$

24. Suppose the periodic signal $x(t)$ has fundamental period T and Fourier coefficients $X[k]$. Let $Y[k]$ be the Fourier coefficient of $y(t)$ where $y(t) = dx(t)/dt$. The Fourier coefficient $X[k]$ will be

- (A) $\frac{TY[k]}{j2\pi k}, k \neq 0$ (B) $\frac{TY[k]}{j2\pi k}$
 (C) $\frac{TY[k]}{jk}, k \neq 0$ (D) $\frac{TY[k]}{jk}$

25. Suppose we have given the following information about a signal $x(t)$:

1. $x(t)$ is real and odd.
2. $x(t)$ is periodic with $T=2$
3. Fourier coefficients $X[k]=0$ for $|k|>1$
4. $\frac{1}{2} \int_0^2 |x(t)|^2 dt = 1$

The signal, that satisfy these condition, is

- (A) $\sqrt{2} \sin \pi t$ and unique
- (B) $\sqrt{2} \sin \pi t$ but not unique
- (C) $2 \sin \pi t$ and unique
- (D) $2 \sin \pi t$ but not unique

26. Consider a continuous-time LTI system whose frequency response is

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt = \frac{\sin 4\omega}{\omega}$$

The input to this system is a periodic signal

$$x(t) = \begin{cases} 2, & 0 \leq t \leq 4 \\ -2, & 4 \leq t \leq 8 \end{cases}$$

with period $T=8$. The output $y(t)$ will be

- (A) $1 + \sin^2 \left(\frac{\pi t}{4} \right)$
- (B) $1 + \cos^2 \left(\frac{\pi t}{4} \right)$
- (C) $1 + \sin \left(\frac{\pi t}{4} \right) + \cos \left(\frac{\pi t}{4} \right)$
- (D) 0

27. Consider a continuous-time ideal low pass filter having the frequency response

$$H(j\omega) = \begin{cases} 1, & |\omega| \leq 80 \\ 0, & |\omega| > 80 \end{cases}$$

When the input to this filter is a signal $x(t)$ with fundamental frequency $\omega_0=12$ and Fourier series coefficients $X[k]$, it is found that $x(t) \xrightarrow{S} y(t) = x(t)$. The largest value of $|k|$, for which $X[k]$ is nonzero, is

- (A) 6
- (B) 80
- (C) 7
- (D) 12

28. A continuous-time periodic signal has a fundamental period $T=8$. The nonzero Fourier series coefficients are as,

$$X[1] = X^*[-1] = j, X[5] = X[-5] = 2,$$

The signal will be

- (A) $4 \cos \left(\frac{\pi}{4} t \right) - 2 \sin \left(\frac{\pi}{4} t \right)$
- (B) $2 \cos \left(\frac{\pi}{4} t \right) + 4 \sin \left(\frac{\pi}{4} t \right)$
- (C) $2 \cos \left(\frac{\pi}{4} t \right) + 2 \sin \left(\frac{\pi}{4} t \right)$
- (D) None of the above

Statement for Q.29-31:

Consider the following three continuous-time signals with a fundamental period of $T=1$

$$x(t) = \cos 2\pi t, y(t) = \sin 2\pi t, z(t) = x(t)y(t)$$

29. The Fourier series coefficient $X[k]$ of $x(t)$ are

- (A) $\frac{1}{2}(\delta[k+1] + \delta[k-1])$
- (B) $\frac{1}{2}(\delta[k+1] - \delta[k-1])$
- (C) $\frac{1}{2}(\delta[k-1] - \delta[k+1])$
- (D) None of the above

30. The Fourier series coefficient of $y(t)$, $Y[k]$ will be

- (A) $\frac{j}{2}(\delta[k+1] + \delta[k-1])$
- (B) $\frac{j}{2}(\delta[k+1] - \delta[k-1])$
- (C) $\frac{j}{2}(\delta[k-1] - \delta[k+1])$
- (D) $\frac{1}{2j}(\delta[k+1] + \delta[k-1])$

31. The Fourier series coefficient of $z(t)$, $Z[k]$ will be

- (A) $\frac{1}{4j}(\delta[k-2] - \delta[k+2])$
- (B) $\frac{1}{2j}(\delta[k-2] - \delta[k+2])$
- (C) $\frac{1}{2j}\delta[k+2] - \delta[k-2]$
- (D) None of the above

32. Consider a periodic signal $x(t)$ whose Fourier series coefficients are

$$X[k] = \begin{cases} 2, & k=0 \\ j \left(\frac{1}{2} \right)^{|k|}, & \text{otherwise} \end{cases}$$

Consider the statements

1. $x(t)$ is real.
2. $x(t)$ is even
3. $\frac{dx(t)}{dt}$ is even

The true statements are

- (A) 1 and 2
- (B) only 2
- (C) only 1
- (D) 1 and 3

Statement for Q.33-36:

A waveform for one period is depicted in figure in question. Determine the trigonometric Fourier series and choose correct option.

33.

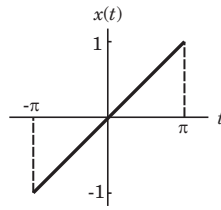


Fig. P5.7.33

- (A) $\frac{2}{\pi}(\cos t + \frac{1}{2}\cos 2t + \frac{1}{3}\cos 3t + \frac{1}{4}\cos 4t + \dots)$
- (B) $\frac{2}{\pi}(\sin t - \frac{1}{2}\sin 2t + \frac{1}{3}\sin 3t - \frac{1}{4}\sin 4t + \dots)$
- (C) $\frac{2}{\pi}(\sin t + \cos t - \frac{1}{2}\sin 2t - \frac{1}{2}\cos 2t + \frac{1}{3}\sin 3t + \dots)$
- (D) $\frac{2}{\pi}(\sin t + \cos t + \frac{1}{3}\sin 3t + \frac{1}{3}\cos 3t + \frac{1}{5}\sin 5t + \dots)$

34.

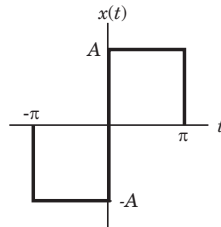


Fig. P5.7.34

- (A) $\frac{A}{2} + \frac{4A}{\pi} \left(\sin t + \frac{1}{2}\sin 2t + \frac{1}{3}\sin 3t + \dots \right)$
- (B) $\frac{A}{2} + \frac{4A}{\pi} \left(\cos t + \frac{1}{3}\cos 3t + \frac{1}{5}\cos 5t + \dots \right)$
- (C) $\frac{4A}{\pi} \left(\sin t + \frac{1}{3}\sin 3t + \frac{1}{5}\sin 5t + \dots \right)$
- (D) $\frac{4A}{\pi} \left(\cos t + \frac{1}{2}\cos 2t + \frac{1}{3}\cos 3t + \dots \right)$

35.

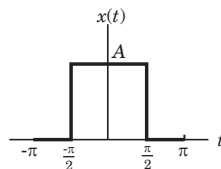


Fig. P5.7.35

- (A) $\frac{A}{2} + \frac{2A}{\pi}(\sin t - \frac{1}{3}\sin 3t + \frac{1}{5}\sin 5t + \dots)$
- (B) $\frac{A}{2} + \frac{2A}{\pi}(\cos t - \frac{1}{2}\cos 2t + \frac{1}{3}\cos 3t + \dots)$
- (C) $\frac{A}{2} + \frac{2A}{\pi}(\cos t - \frac{1}{3}\cos 3t + \frac{1}{5}\cos 5t + \dots)$
- (D) $\frac{A}{2} + \frac{2A}{\pi}(\sin t + \cos t + \frac{1}{3}\sin 3t + \frac{1}{3}\cos 3t + \dots)$

36.

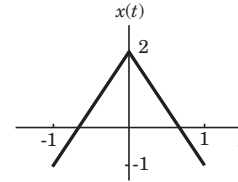


Fig. P5.7.36

- (A) $\frac{1}{2} + \frac{12}{\pi^2}(\cos \pi t + \frac{1}{9}\cos 3\pi t + \frac{1}{25}\cos 5\pi t + \dots)$
- (B) $3 + \frac{12}{\pi^2}(\cos \pi t + \frac{1}{9}\cos 3\pi t + \frac{1}{25}\cos 5\pi t + \dots)$
- (C) $\frac{1}{2} + \frac{12}{\pi^2}(\sin \pi t - \frac{1}{9}\sin 3\pi t + \frac{1}{25}\sin 5\pi t - \dots)$
- (D) $3 + \frac{12}{\pi^2}(\sin \pi t - \frac{1}{9}\sin 3\pi t + \frac{1}{25}\sin 5\pi t - \dots)$

SOLUTIONS

$$1. (D) X[k] = \frac{1}{T} \int_{-T/2}^{T/2} A \delta(t) e^{-jk\omega_0 t} dt = \frac{A}{T},$$

$$A = 10, T = 5, X[k] = 2$$

$$2. (C) X[k] = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-T/4}^{T/4} A e^{-jk\omega_0 t} dt$$

$$= \frac{A}{T} \left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0} \right]_{-T/4}^{T/4} = \frac{A}{\pi k} \sin\left(\frac{\pi k}{2}\right)$$

$$3. (B) T = 2\pi, \omega_0 = \frac{2\pi}{2\pi} = 1, x(t) = \begin{cases} A, & 0 < t < \frac{4\pi}{3} \\ 0, & \frac{4\pi}{3} < t < 2\pi \end{cases}$$

$$X[k] = \frac{1}{2\pi} \int_0^{2\pi} x(t) e^{-jkt} dt = \frac{1}{2\pi} \int_0^{4\pi/3} A e^{-jkt} dt = \frac{jA}{2\pi k} \left[e^{-j\left(\frac{4\pi k}{3}\right)} - 1 \right]$$

$$4. (C) T = 2, \omega_0 = \frac{2\pi}{2} = \pi, x(t) = \begin{cases} -A, & -1 < t < 0 \\ A, & 0 < t < 1 \end{cases}$$

$$X[k] = \frac{1}{2} \int_{-1}^1 x(t) e^{-jkt} dt = \frac{1}{2} \left(\int_{-1}^0 -A e^{-jkt} dt + \int_0^1 A e^{-jkt} dt \right)$$

$$= \frac{A}{2} \left(\frac{1 - e^{jk\pi}}{jk\pi} + \frac{e^{-jk\pi} - 1}{-jk\pi} \right) = \frac{A}{jk\pi} (1 - (-1)^k)$$

$$5. (A) \sin^2 t = \left(\frac{e^{jt} - e^{-jt}}{2j} \right)^2 = \frac{-1}{4} (e^{2jt} - 2 + e^{-2jt})$$

$$\text{The fundamental period of } \sin^2(t) \text{ is } \pi \text{ and } \omega_0 = \frac{2\pi}{\pi} = 2,$$

$$X[k] = \frac{-1}{4} \delta[k-1] + \frac{1}{2} \delta[k] - \frac{1}{4} \delta[k+1]$$

$$7. (C) x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{j2\pi kt} = j e^{j2\pi t} - j e^{-j2\pi t} + e^{j6\pi t} + e^{-j6\pi t}$$

$$= -2 \sin 2\pi t + 2 \cos 6\pi t$$

$$8. (D) x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{jkt} = \sum_{k=-\infty}^{-1} \left(\frac{-1}{3} \right)^{-k} e^{jkt} + \sum_{k=0}^{\infty} \left(\frac{-1}{3} \right)^k e^{jkt}$$

$$= \frac{-1}{3} e^{-jt} + \frac{1}{1 + \frac{1}{3} e^{-jt}} = \frac{4}{5 + 3 \cos t}$$

$$9. (D) x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{j\pi kt}$$

$$= 2e^{-j\frac{\pi}{4}} e^{j(-4)\pi t} + e^{j\frac{\pi}{4}} e^{j(-3)\pi t} + e^{-j\frac{\pi}{4}} e^{j(3)\pi t} + 2e^{j\frac{\pi}{4}} e^{j(4)\pi t}$$

$$= 2(e^{-j(4\pi t + \pi/4)} + e^{j(4\pi t + \pi/4)}) + (e^{-j(3\pi t - \pi/4)} + e^{j(3\pi t - \pi/4)})$$

$$= 4 \cos(4\pi t + \pi/4) + 2 \cos(3\pi t - \pi/4)$$

$$10. (A) X[k] = e^{-j2\pi k}, -4 \leq k \leq 4$$

$$x(t) = \sum_{k=-4}^4 e^{-j2\pi k} e^{j\pi kt} = \sum_{k=-4}^4 e^{-j2\pi k(t-1)} = \frac{\sin 9\pi t}{\sin \pi t}$$

$$11. (D) X[k] = |k|, -3 \leq k \leq 3$$

$$x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{j\pi kt}$$

$$= 3e^{j(-3)\pi t} + 2e^{j(-2)\pi t} + e^{j(-1)\pi t} + e^{j(1)\pi t} + 2e^{j(2)\pi t} + 3e^{j(3)\pi t}$$

$$= 6 \cos 3\pi t + 4 \cos 2\pi t + 2 \cos \pi t$$

12. (A) $x(t - t_0)$ is also periodic with T . The Fourier series coefficients $X_1[k]$ of $x(t - t_0)$ are

$$X_1[k] = \frac{1}{T} \int_T x(t - t_0) e^{-jk\omega_0 t} dt = \frac{e^{-jk\omega_0 t_0}}{T} \int_T x(\tau) e^{-jk\omega_0 \tau} d\tau$$

$$= e^{-jk\omega_0 t_0} X[k]$$

Similarly, the FS coefficients of $x(t + t_0)$ are

$$X_2[k] = e^{jk\omega_0 t_0} X[k]$$

The FS coefficients of $x(t - t_0) + x(t + t_0)$ are

$$Y[k] = X_1[k] + X_2[k] = e^{-jk\omega_0 t_0} X[k] + e^{jk\omega_0 t_0} X[k]$$

$$= 2 \cos(\omega_0 k t_0) X[k]$$

$$13. (A) \text{Ev}\{x(t)\} = \frac{x(t) + x(-t)}{2},$$

The FS coefficients of $x(t)$ are

$$X_1[k] = \frac{1}{T} \int_T x(-t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(\tau) e^{jk\omega_0 \tau} d\tau = X[-k]$$

Therefore, the FS coefficients of $\text{Ev}\{x(t)\}$ are

$$Y[k] = \frac{X[k] + X[-k]}{2}$$

$$14. (C) \text{Re}\{x(t)\} = \frac{x(t) + x^*(t)}{2},$$

The FS coefficient of $x^*(t)$ is

$$X_1[k] = \frac{1}{T} \int_T x^*(t) e^{-jk\omega_0 t} dt = X_1^*[-k]$$

$$X_1^*[k] = \frac{1}{T} \int_T x(t) e^{jk\omega_0 t} dt = X[-k]$$

$$X_1[k] = X^*[-k]$$

$$Y[k] = \frac{X[k] + X^*[-k]}{2}$$

13. $X[k] = \cos\left(\frac{10\pi}{19}k\right) + 2j \sin\left(\frac{4\pi}{19}k\right)$

(A) $\frac{19}{2}(\delta[n+5] + \delta[n-5]) + 19(\delta[n+2] - \delta[n-2]), |n| \leq 9$

(B) $\frac{1}{2}(\delta[n+5] + \delta[n-5]) + (\delta[n+2] - \delta[n-2]), |n| \leq 9$

(C) $\frac{9}{2}(\delta[n+5] + \delta[n-5]) + 9(\delta[n+2] - \delta[n-2]), |n| \leq 9$

(D) $\frac{1}{2}(\delta[n+5] + \delta[n-5]) + (\delta[n+2] - \delta[n-2]), |n| \leq 9$

14. $X[k] = \cos\left(\frac{\pi k}{21}\right)$

(A) $\frac{21}{2}(\delta[n+4] + \delta[n-4]), |n| \leq 10$

(B) $\frac{1}{2}(\delta[n+4] + \delta[n-4]), |n| \leq 10$

(C) $\frac{21}{2}(\delta[n+4] - \delta[n-4]), |n| \leq 10$

(D) $\frac{1}{2}(\delta[n+4] - \delta[n-4]), |n| \leq 10$

Statement for Q.15-20:

Consider a periodic signal $x[n]$ with period N and FS coefficients $X[k]$. Determine the FS coefficients $Y[k]$ of the signal $y[n]$ given in question.

15. $y[n] = x[n - n_0]$

(A) $e^{j\left(\frac{2\pi}{N}\right)n_0 k} X[k]$ (B) $e^{-j\left(\frac{2\pi}{N}\right)n_0 k} X[k]$

(C) $kX[k]$ (D) $-kX[k]$

16. $y[n] = x[n] - x[n-2]$

(A) $\sin\left(\frac{4\pi}{N}k\right) X[k]$ (B) $\cos\left(\frac{4\pi}{N}k\right) X[k]$

(C) $\left(1 - e^{-j\left(\frac{4\pi}{N}\right)k}\right) X[k]$ (D) $\left(1 - e^{j\left(\frac{4\pi}{N}\right)k}\right) X[k]$

17. $y[n] = x[n] + x[n + N/2]$, (assume that N is even)

(A) $2X[2k-1]$, for $0 \leq k \leq \left(\frac{N}{2} - 1\right)$

(B) $2X[2k-1]$, for $0 \leq k \leq \frac{N}{2}$

(C) $2X[2k]$, for $0 \leq k \leq \left(\frac{N}{2} - 1\right)$

(D) $2X[2k]$, for $0 \leq k \leq \frac{N}{2}$

18. $y[n] = x[n] - x[n - N/2]$, (assume that N is even)

(A) $(1 - (-1)^{k+1})X[2k]$ (B) $(1 - (-1)^k)X[k]$

(C) $(1 - (-1)^{k+1})X[k]$ (D) $(1 - (-1)^k)X[2k]$

19. $y[n] = x^*[-n]$

(A) $-X^*[k]$ (B) $-X^*[-k]$

(C) $X^*[k]$ (D) $X^*[-k]$

20. $y[n] = (-1)^n x[n]$, (assume that N is even)

(A) $X\left[k - \frac{N}{2}\right]$ (B) $X\left[k + \frac{N}{2}\right]$

(C) $X\left[k - \frac{N}{2} + 1\right]$ (D) $X\left[k + \frac{N}{2} - 1\right]$

Statement for Q.21-23:

Consider a discrete-time periodic signal

$$x[n] = \begin{cases} 1, & 0 \leq n \leq 7 \\ 0, & 8 \leq n \leq 9 \end{cases}$$

with period $N = 10$. Also $y[n] = x[n] - x[n-1]$

21. The fundamental period of $y[n]$ is

(A) 9 (B) 10

(C) 11 (D) None of the above

22. The FS coefficients of $y[n]$ are

(A) $\frac{1}{10} \left(1 - e^{j\left(\frac{8\pi}{5}\right)k}\right)$ (B) $\frac{1}{10} \left(1 - e^{-j\left(\frac{8\pi}{5}\right)k}\right)$

(C) $\frac{1}{10} \left(1 - e^{j\left(\frac{4\pi}{5}\right)k}\right)$ (D) $\frac{1}{10} \left(1 - e^{-j\left(\frac{4\pi}{5}\right)k}\right)$

23. The FS coefficients of $x[n]$ are

(A) $-\frac{j}{2} e^{-j\left(\frac{\pi k}{10}\right)} \operatorname{cosec}\left(\frac{\pi k}{10}\right) Y[k], k \neq 0$

(B) $\frac{j}{2} e^{-j\left(\frac{\pi k}{10}\right)} \operatorname{cosec}\left(\frac{\pi k}{10}\right) Y[k], k \neq 0$

(C) $-\frac{1}{2} e^{-j\left(\frac{\pi k}{10}\right)} \sec\left(\frac{\pi k}{10}\right) Y[k]$

(D) $\frac{1}{2} e^{-j\left(\frac{\pi k}{10}\right)} \sec\left(\frac{\pi k}{10}\right) Y[k]$

Statement for Q.24-27:

Consider a discrete-time signal with Fourier representation.

$$x[n] \xleftrightarrow{DTFS: \frac{\pi}{10}} X[k]$$

In question the FS coefficient $Y[k]$ is given. Determine the corresponding signal $y[n]$ and choose correct option.

24. $Y[k] = X[k-5] + X[k+5]$

(A) $2 \sin\left(\frac{\pi}{5}n\right)x[n]$ (B) $2 \cos\left(\frac{\pi}{5}n\right)x[n]$

(C) $2 \sin\left(\frac{\pi}{2}n\right)x[n]$ (D) $2 \cos\left(\frac{\pi}{2}n\right)x[n]$

25. $Y[k] = \cos\left(\frac{\pi k}{5}\right)X[k]$

(A) $\frac{1}{2}(x[n+5] + x[n-5])$ (B) $\frac{1}{2}(x[n+2] + x[n-2])$

(C) $\frac{1}{2}(x[n+10] + x[n-10])$ (D) None of the above

26. $Y[k] = X[k] * X[k]$

(A) $\frac{(x[n])^2}{2\pi}$ (B) $j2\pi(x[n])^2$

(C) $(x[n])^2$ (D) $2\pi(x[n])^2$

27. $Y[k] = \text{Re}\{X[k]\}$

(A) $\frac{x[n] + x[-n]}{2}$ (B) $\frac{x[n] - x[-n]}{2}$

(C) $\frac{x[n] - x[-n]}{2\pi}$ (D) $\frac{x[n] + x[-n]}{2\pi}$

28. Consider a sequence $x[n]$ with following facts :

1. $x[n]$ is periodic with $N = 6$

2. $\sum_{n=0}^5 x[n] = 2$

3. $\sum_{n=-2}^7 (-1)^n x[n] = 1$

4. $x[n]$ has the minimum power per period among the set of signals satisfying the preceding three condition.

The sequence would be..

(A) $\left\{\dots, \frac{1}{2}, \frac{1}{6}, \frac{1}{2}, \frac{1}{6}, \frac{1}{2}, \dots\right\}$ (B) $\left\{\dots, 0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$

(C) $\left\{\dots, \frac{1}{3}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6}, \frac{1}{3}, \dots\right\}$ (D) $\{..0, 1, 2, 3, 4, \dots\}$

29. A real and odd periodic signal $x[n]$ has fundamental period $N = 7$ and FS coefficients $X[k]$. Given that $X[15] = j$, $X[16] = 2j$, $X[17] = 3j$. The values of $X[0]$, $X[-1]$, $X[-2]$, and $X[-3]$ will be

(A) $0, j, 2j, 3j$ (B) $1, 1, 2, 3$
(C) $1, -1, -2, -3$ (D) $0, -j, -2j, -3j$

30. Consider a signal $x[n]$ with following facts

1. $x[n]$ is a real and even signal

2. The period of $x[n]$ is $N = 10$

3. $X[11] = 5$

4. $\frac{1}{10} \sum_{n=0}^9 |X[k]|^2 = 50$

The signal $x[n]$ is

(A) $5 \cos\left(\frac{\pi}{10}n\right)$ (B) $5 \sin\left(\frac{\pi}{10}n\right)$

(C) $10 \cos\left(\frac{\pi}{5}n\right)$ (D) $10 \sin\left(\frac{\pi}{5}n\right)$

31. Each of two sequence $x[n]$ and $y[n]$ has a period $N = 4$. The FS coefficient are

$$X[0] = X[3] = \frac{1}{2}, X[1] = \frac{1}{2}, X[2] = 1 \text{ and}$$

$$Y[0], Y[1], Y[2], Y[3] = 1$$

The FS coefficient $Z[k]$ for the signal $z[n] = x[n]y[n]$ will be

(A) 6 (B) $6|k|$

(C) $6^{|k|}$ (D) $e^{j\frac{\pi}{2}k}$

32. Consider a discrete-time periodic signal

$$x[n] = \frac{\sin\left(\frac{11\pi}{20}n\right)}{\sin\left(\frac{\pi}{20}n\right)}$$

with a fundamental period $N = 20$. The Fourier series coefficients of this function are

(A) $\frac{1}{20}(u[k+5] - u[k-6]), |k| \leq 10$

(B) $\frac{1}{20}(u[k+5] - u[k-5]), |k| \leq 10$

(C) $(u[k+5] - u[k-6]), |k| \leq 10$

(D) $(u[k+5] - u[k-6]), |k| \leq 10$

11. (D) $N=7, \Omega_o = \frac{2\pi}{7},$

$$x[n] = \sum_{n=-3}^3 X[k] e^{j\left(\frac{2\pi}{7}\right)kn} = 2e^{j(-1)\left(\frac{2\pi}{7}\right)n} - 1 + 2e^{j(1)\left(\frac{2\pi}{7}\right)n}$$

$$= 4 \cos\left(\frac{2\pi}{7}n\right) - 1$$

12. (C) $N=12, \Omega_o = \frac{\pi}{6}, X[k] = e^{-j\left(\frac{\pi}{6}\right)k}$

$$x[n] = \sum_{k=-6}^6 e^{-j\left(\frac{\pi}{6}\right)k} e^{j\left(\frac{\pi}{6}\right)kn} = \sum_{k=-6}^6 e^{j\left(\frac{\pi}{6}\right)k(n-1)}$$

$$= \frac{e^{j(-4)\frac{\pi}{6}(n-1)} \left(1 - e^{j\frac{9\pi}{6}(n-1)}\right)}{\left(1 - e^{j\frac{\pi}{6}(n-1)}\right)} = \frac{\sin\left(\frac{3\pi}{4}(n-1)\right)}{\sin\left(\frac{\pi}{12}(n-1)\right)}$$

13. (A) $N=19, \Omega_o = \frac{2\pi}{19}$

$$X[k] = \cos\left(\frac{10\pi}{19}k\right) + 2j \sin\left(\frac{10\pi}{19}k\right)$$

$$= \frac{1}{2} \left(e^{-j(-5)\frac{2\pi}{19}k} + e^{-j(5)\frac{2\pi}{19}k} \right) + \left(e^{-j(-2)\frac{2\pi}{19}k} + e^{-j(2)\frac{2\pi}{19}k} \right)$$

By inspection

$$x[n] = \frac{19}{2} (\delta[n+5] + \delta[n-5]) + 19 (\delta[n+2] + \delta[n-2]),$$

Where $|n| \leq 9$

14. (A) $N=21, \Omega_o = \frac{2\pi}{21}$

$$X[k] = \cos\left(\frac{8\pi}{21}k\right) = \frac{1}{2} \left(e^{-j(-4)\frac{2\pi}{21}k} + e^{-j(4)\frac{2\pi}{21}k} \right)$$

Since $X[k] = \frac{1}{N} \sum_{n=N} x[n] e^{-jk\Omega_o n}$, By inspection

$$x[n] = \begin{cases} \frac{21}{2}, & n = \pm 4 \\ 0, & \text{otherwise } n \in \{-10, -9, \dots, 9, 10\} \end{cases}$$

15. (B) $Y[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n - n_o] e^{-j\left(\frac{2\pi}{N}\right)kn}$

$$= \frac{1}{N} e^{-j\left(\frac{2\pi}{N}\right)kn_o} \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn} = e^{-j\left(\frac{2\pi}{N}\right)kn_o} X[k]$$

16. (C) $Y[k] = X[k] - e^{-j\left(\frac{2\pi}{N}\right)2k} X[k] = \left(1 - e^{-j\left(\frac{4\pi}{N}\right)k}\right) X[k]$

17. (C) Note that $y[n] = x[n] + x[n + N/2]$ has a period of $N/2$ and N has been assumed to be even,

$$Y[k] = \frac{2}{N} \sum_{n=0}^{N/2-1} (x[n] + x[n + N/2]) e^{-j\left(\frac{4\pi}{N}\right)kn}$$

$$= 2X[2k] \text{ for } 0 \leq k \leq (N/2 - 1)$$

18. (B) $y[n] = x[n] - x[n - N/2]$

$$Y[k] = \left(1 - e^{-j\left(\frac{2\pi}{N}\right)\frac{N}{2}k}\right) X[k] = (1 - e^{-j\pi k}) X[k]$$

$$= \begin{cases} 0, & k \text{ even} \\ 2X[k], & k \text{ odd} \end{cases}$$

19. (C) $y[n] = x^*[-n]$

$$Y[k] = \frac{1}{N} \sum_{n=0}^{N-1} x^*[-n] e^{-j\left(\frac{2\pi}{N}\right)kn} = X^*[k]$$

20. (A) With N even

$$y[n] = (-1)^n x[n] = e^{j\pi n} x[n] = e^{j\left(\frac{2\pi}{N}\right)\frac{N}{2}n} x[n]$$

$$Y[k] = \frac{1}{N} \sum_{n=0}^{N-1} e^{j\left(\frac{2\pi}{N}\right)\frac{N}{2}n} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn}$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)n\left(k - \frac{N}{2}\right)} = X[k - N/2]$$

21. (B) $y[n]$ is shown in fig. S5.8.21. It has fundamental



Fig. S5.8.21

period of 10.

22. (B) $Y[k] = \frac{1}{10} \sum_{n=0}^9 y[n] e^{-j\left(\frac{2\pi}{10}\right)kn}$

$$= \frac{1}{10} \left(1 - e^{-j\left(\frac{2\pi}{10}\right)k8}\right) = \frac{1}{10} \left(1 - e^{-j\left(\frac{8\pi}{5}\right)k}\right)$$

23. (A) $y[n] = x[n] - x[n - 1]$

$$Y[k] = X[k] - e^{-j\left(\frac{2\pi}{10}\right)k} X[k] \Rightarrow X[k] = \frac{Y[k]}{1 - e^{-j\left(\frac{\pi}{5}\right)k}}$$

$$\Rightarrow X[k] = \frac{e^{j\left(\frac{\pi}{10}\right)k} Y[k]}{e^{j\left(\frac{\pi}{10}\right)k} - e^{-j\left(\frac{\pi}{10}\right)k}} = \frac{e^{j\left(\frac{\pi}{10}\right)k} Y[k]}{2j \sin\left(\frac{\pi k}{10}\right)}$$

$$= \frac{-j}{2} e^{-j\left(\frac{\pi}{10}\right)k} \operatorname{cosec}\left(\frac{\pi}{10} k\right) Y[k]$$

24. (D) $\Omega_o = \frac{\pi}{10}$, $Y[k] = X[k-5] + X[k+5]$

$$\Rightarrow y[n] = \left(e^{j(5)\frac{\pi}{10}n} + e^{j(-5)\frac{\pi}{10}n} \right) x[n] = 2 \cos\left(\frac{\pi}{2} n\right) x[n]$$

25. (B) $Y[k] = \cos\left(\frac{\pi}{5} k\right) X[k] = \left(\frac{e^{j\frac{\pi}{5}k} + e^{-j\frac{\pi}{5}k}}{2} \right) X[k]$

$$= \frac{1}{2} \left(e^{j(2)\frac{\pi}{10}k} + e^{j(-2)\frac{\pi}{10}k} \right) X[k]$$

$$\Rightarrow y[n] = \frac{1}{2} (x[n-2] + x[n+2])$$

26. (C) $Y[k] = X[k] * X[k] \Rightarrow y[n] = x[n]x[n] = (x[n])^2$

27. (A) $Y[k] = \operatorname{Re}\{X[k]\} \Rightarrow y[n] = \operatorname{Ev}\{x[n]\} = \frac{x[n] + x[-n]}{2}$

28. (A) $N = 6$, $\Omega_o = \frac{2\pi}{6}$,

From fact 2, $\sum_{n=0}^5 x[n] = 2$

$$\Rightarrow \frac{1}{6} \sum_{n=0}^5 e^{j\left(\frac{2\pi}{6}\right)(0)k} x[n] = \frac{1}{3} \Rightarrow X[0] = \frac{1}{3},$$

From fact 3, $\sum_{n=2}^7 (-1)^n x[n] = 1$

$$\Rightarrow \frac{1}{6} \sum_{n=0}^5 e^{j\left(\frac{2\pi}{6}\right)(3)k} x[n] = \frac{1}{6}, \quad X[3] = \frac{1}{6}$$

By Parseval's relation, the average power in $x[n]$ is

$$P = \sum_{k=0}^5 |X[k]|^2,$$

The value of P is minimized by choosing

$$X[1] = X[2] = X[4] = X[5] = 0$$

Therefore

$$x[n] = X[0] + X[3]e^{j\left(\frac{2\pi}{6}\right)3n} = \frac{1}{3} + (-1)^n \frac{1}{6} = \frac{1}{3} + (-1)^n \frac{1}{6}$$

$$x[n] = \left\{ \dots, \frac{1}{2}, \frac{1}{6}, \frac{1}{2}, \frac{1}{6}, \frac{1}{2}, \dots \right\}$$

↑

29. (D) Since the FS coefficient repeat every N . Thus

$$X[1] = X[15], \quad X[2] = X[16], \quad X[3] = X[17]$$

The signal real and odd, the FS coefficient $X[k]$ will be purely imaginary and odd. Therefore $X[0] = 0$

$$X[-1] = -X[1], \quad X[-2] = -X[2], \quad X[-3] = -X[3]$$

Therefore (D) is correct option.

30. (C) Since $N = 10$, $X[11] = X[1] = 5$

Since $x[n]$ is real and even $X[k]$ is also real and even.

$$\text{Therefore } X[1] = X[-1] = 5.$$

$$\text{Using Parseval's relation } \sum_N |X[k]|^2 = 50 = \sum_{k=-1}^8 |X[k]|^2$$

$$|X[-1]|^2 + |X[1]|^2 + |X[0]|^2 + \sum_{k=2}^8 |X[k]|^2 = 50$$

$$|X[0]|^2 + \sum_{k=2}^8 |X[k]|^2 = 0$$

Therefore $X[k] = 0$ for $k = 0, 2, 3, \dots, 8$.

$$x[n] = \sum_N X[k] e^{j\left(\frac{2\pi}{N}\right)kn} = \sum_{k=-1}^8 X[k] e^{j\left(\frac{2\pi}{10}\right)kn}$$

$$= 5 \left(e^{-j\left(\frac{2\pi}{10}\right)n} + e^{j\left(\frac{2\pi}{10}\right)n} \right) = 10 \cos\left(\frac{\pi}{5} n\right)$$

31. (A) $z[n] = x[n]y[n] \xrightarrow{DTFS} \sum_{k=-N}^N X[l]Y[k-l]$

$$\Rightarrow Z[k] = \sum_{l=0}^3 X[l]Y[k-l]$$

$$\Rightarrow Z[k] = X[0]Y[k] + X[1]Y[k-1] + X[2]Y[k-2] + X[3]Y[k-3]$$

$$= Y[k] + 2Y[k-1] + 2Y[k-2] + Y[k-3]$$

Since $Y[k]$ is 1 for all values of k .

Thus $Z[k] = 6$, for all k .

32. (A) $N = 20$ We know that

$$\begin{cases} 1, & |n| \leq 5 \\ 0, & 5 < |n| \leq 10 \end{cases} \xleftrightarrow{DTFS, \frac{\pi}{10}} \frac{\sin\left(\frac{11\pi}{20} k\right)}{\sin\left(\frac{\pi}{20} k\right)}$$

Using duality

$$\frac{\sin\left(\frac{11\pi}{20} n\right)}{\sin\left(\frac{\pi}{20} n\right)} \xleftrightarrow{DTFS, \frac{\pi}{10}} \frac{1}{20} \begin{cases} 1, & |k| \leq 5 \\ 0, & 5 < |k| \leq 10 \end{cases}$$
